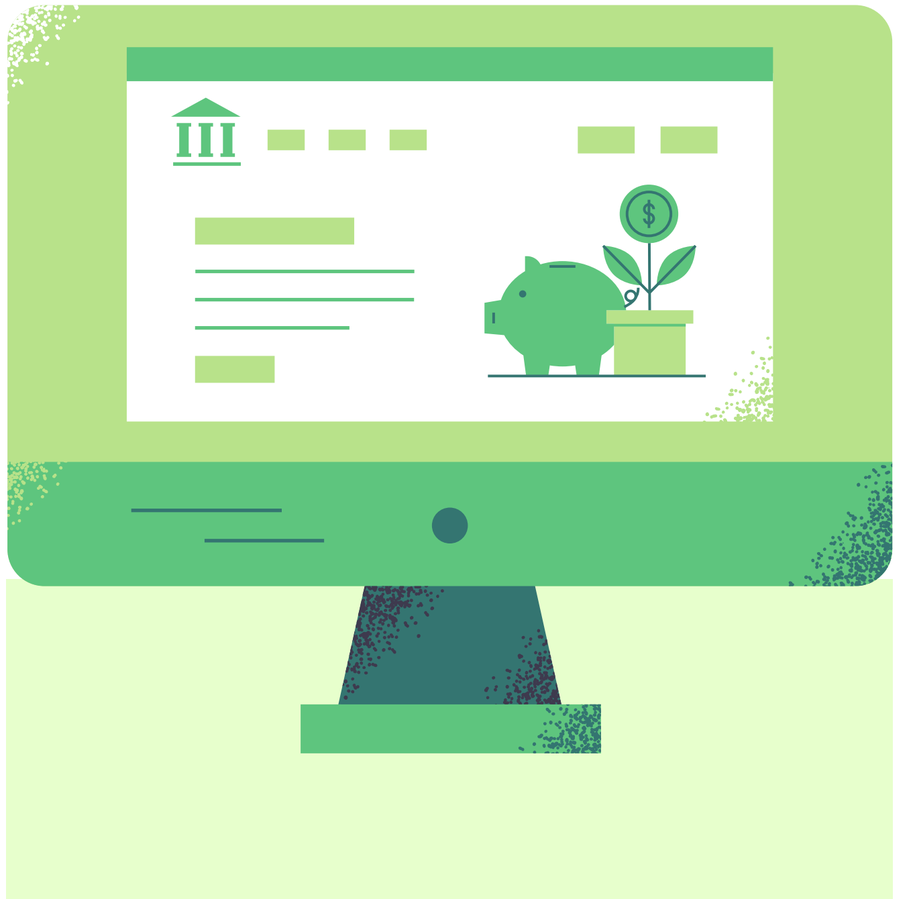


DATABANK

From business case to a running data pipeline
Snowflake on AWS Paris · dbt · GitHub Actions

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From business case to running pipeline



- Business case & use cases
- Architecture
- 6 sources & ingestion
- Transform, serve & proof
- Reliability
- Security, costs & cloud
- Best case & forecast

Databank at a glance



A fictive French mobile bank: current account, debit card and share investing, all in one app.

Every card payment, balance change and market price is data Databank must process, show to its clients within minutes, and report to the Banque de France and Tracfin.

Assumptions in docs/company.md · our dataset is a 4 % sample (10,000 clients)

250 k

clients

80 k

card payments per day

< 2 min

for a card payment to show in the app

€0

spent: free trial credits only

3 use cases

UC1 · Live balance

A card payment shows in the app in under 2 minutes.

Available balance = booked balance minus pending payments.

For: clients

UC2 · Spending & portfolio

Daily spending per client and category.

Share portfolio valued every day in euros, with real ECB exchange rates.

For: clients, advisors

UC3 · Regulator & anti-money laundering

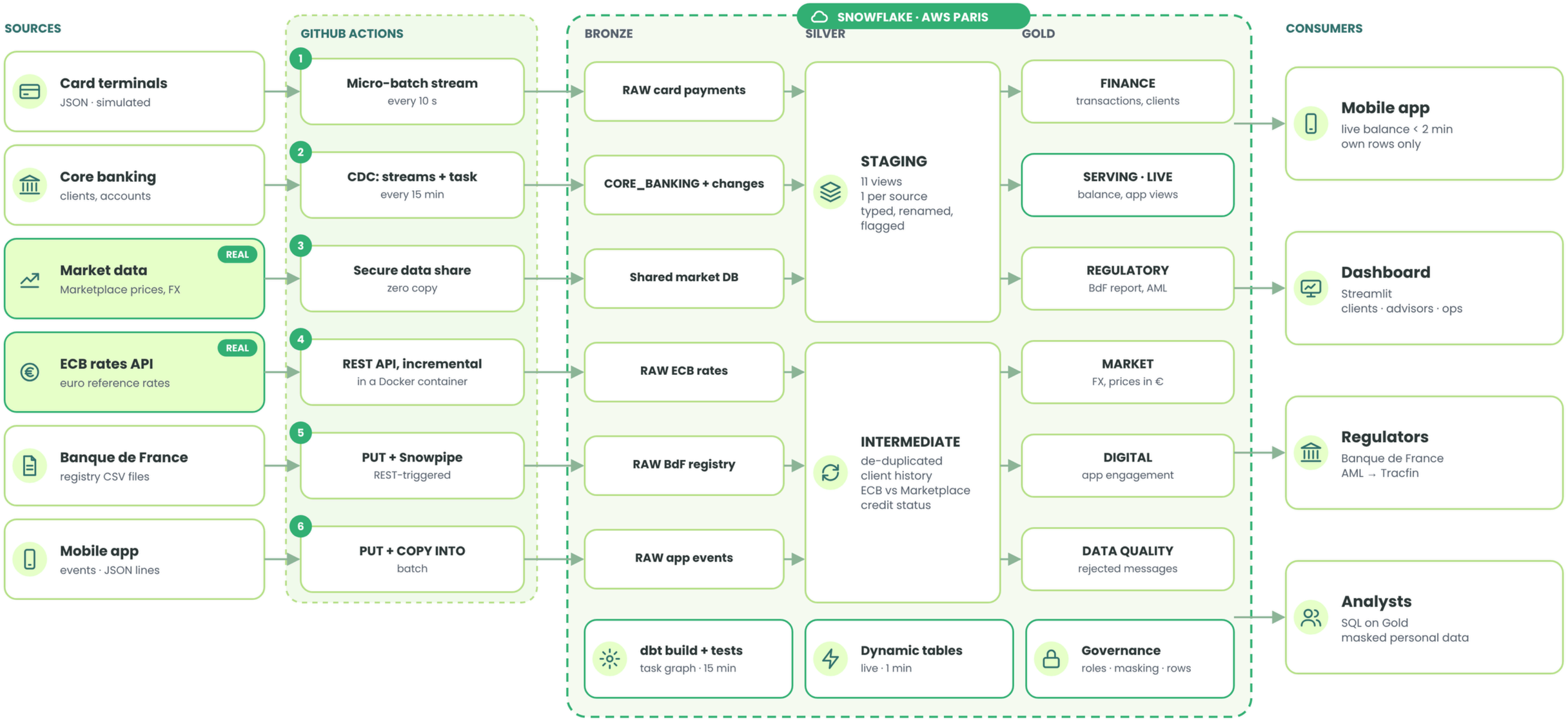
Daily Banque de France card report, ready before 09:00 and reproducible.

Suspicious payments flagged for Tracfin.

For: compliance, Banque de France

Architecture today

Snowflake on AWS Paris · dbt · GitHub Actions



6 sources, 6 ingestion methods

- A · Card terminals: micro-batch stream every 10 s
  B · Core banking: change data capture (streams)
- C · Market data (real): Marketplace share
  D · ECB rates (real): REST API, in Docker
- E · Banque de France files: PUT + Snowpipe
  F · Mobile app events: PUT + COPY INTO



Transform & serve

dbt

32 models and 43 tests, from raw data to Gold.

Runs every 15 minutes in a task graph; a failed test stops what depends on it.

Live path

2 dynamic tables refresh every minute: card payments and live balance.

A payment reaches Gold a few seconds after it is sent.

Serving

Streamlit dashboard for clients, advisors and ops.

The app sees only the logged-in client's rows (row access policy).

Live demo: dashboard, client C000042, pending payments

75 / 75

dbt models and tests pass

301,118

payments in Gold, 0 duplicates

2,036

real ECB rates loaded by the Docker job

Proof it works

The simulator injects errors at known rates, and the pipeline finds exactly those: 0.99 % duplicates, 0.18 % broken amounts, 0.30 % large payments.

The Banque de France figure for 15 Oct, recomputed from raw data without dbt: identical (€176,580.44).

Reliability

A check that fails

A payment dated 2030 is injected. The test fails, the 13 steps after it are skipped, and the regulator report keeps its last good version.

Row removed, re-run: all green.

Incident 1 · Replay

A loader sends 500 payments twice. Raw keeps them for audit; Gold removes duplicates.

Business figures: unchanged.
Owners: Thomas, Adrien

Incident 2 · ECB down

No new ECB rate: the freshness check warns, portfolios are valued with Marketplace rates.

When the API is back, the job catches up alone.
Owner: Thomas

Both incidents tested on 3 Nov 2026 · details in docs/incidents.md

Security & costs

Security: 5 roles, least privilege. Personal data masked for everyone except compliance; each client sees only their own rows. Key pairs, no passwords, no secret in the code.

Costs: €0. About 11.5 of the 400 free trial credits used. A resource monitor caps compute at 20 credits a month, and everything is paused after each demo.



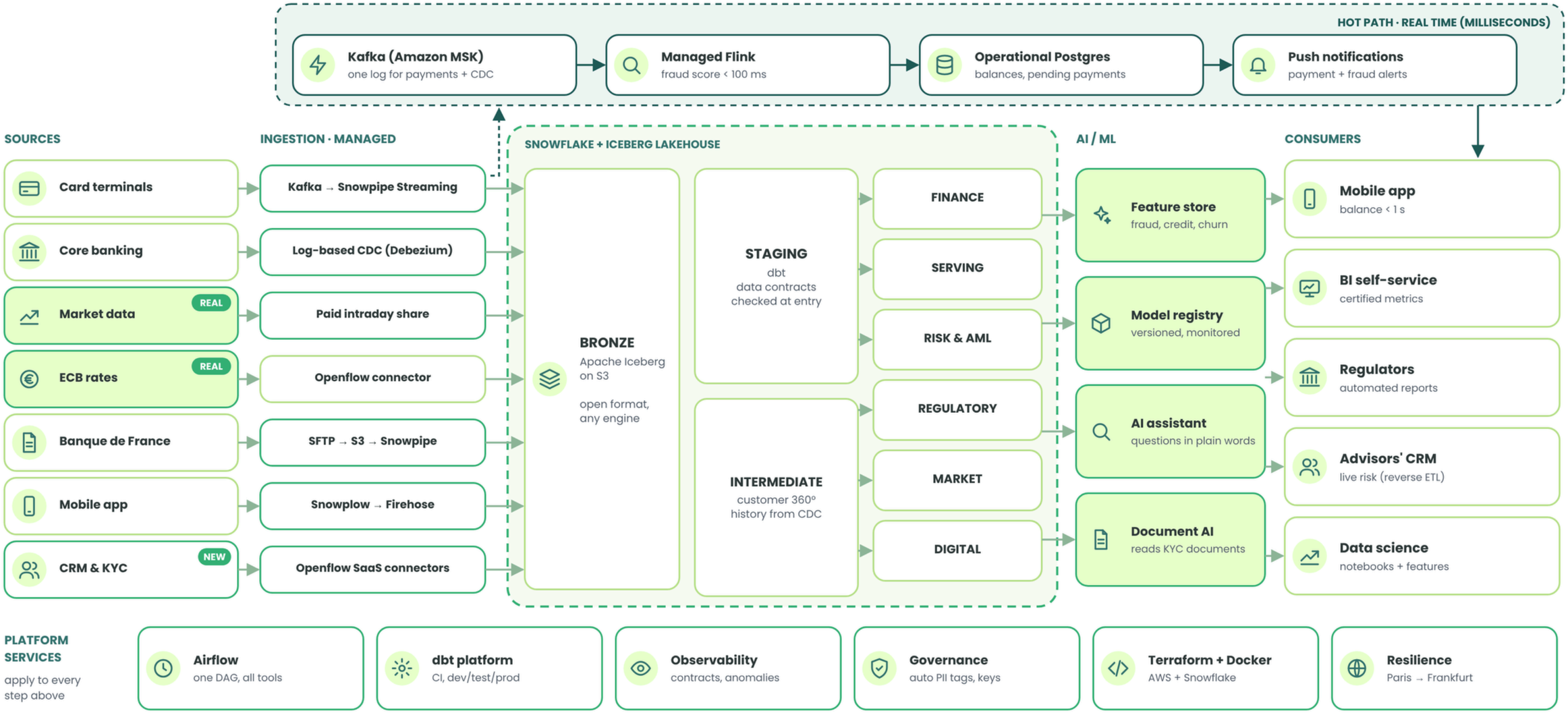


Cloud & IaC

- Terraform creates a warehouse, a role and a schema. Its state is encrypted in Snowflake, versioned, one run at a time.
- GitHub Actions runs the loaders; the ECB job runs in a Docker container, as a non-root user. One cloud: AWS Paris.

Best case

If we had all the budget and the best platforms



Green outline = new or upgraded compared with today · Dark teal = new real-time hot path · Same Bronze / Silver / Gold layers and dbt models as today

Forecast 2026 → 2030

Based on Gartner signals and emerging tools

	TODAY · 2026 · what we built	2027 · foundations	2028 · real time	2029 – 2030 · governed AI
Gartner signals	**Already true** • 40% of business apps embed AI agents by end of 2026 (from < 5% in 2025) • 60% of AI projects without AI-ready data abandoned	**2027** • > 40% of agentic AI projects cancelled (cost, unclear value, weak control) • EU AI Act for credit scoring: 2 December 2027	**2028** • Event-driven data streaming > 60% of companies (from < 15% in 2025) • 15% of daily work decisions made by agents	**2029 – 2030** • Semantic layer = critical infrastructure • 50% use agents to write data contracts • 40% of companies use GraphRAG
Data & streaming	**Today** • Micro-batches every 10 s • 6 ingestion methods • dbt: 32 models, 43 tests	**2027** • Kafka + log-based CDC • Data contracts on every source • Terraform for every object	**2028** • Most flows event-driven • Fraud score before payment • Bronze on Apache Iceberg	**2029 – 2030** • Open lakehouse, several engines • Platform serves agents as well as dashboards
AI & agents	**Today** • Rule-based fraud score • No AI in production	**2027** • AI assistant for analysts, on governed Gold data only • Credit model documented	**2028** • Supervised agents: alert triage, report drafts; a human approves • Metrics defined once	**2029 – 2030** • Graph-based anti-money laundering • Agents take small, reversible decisions within strict limits
Governance & rules	**Today** • 5 roles, masking, row access • EU region only (Paris) • Lineage: file, batch, run id	**2027** • Dev / test / prod environments • EU AI Act ready (credit scoring) • Replica in Frankfurt (DORA)	**2028** • Personal data tagged automatically • Masking applied by tag • Every automated decision logged	**2029 – 2030** • Decision governance in production • Contracts drafted by agents, approved by humans

Grey = Gartner published predictions and the EU AI Act deadline (sources in docs/forecast.md) · Dark outline = what Databank runs today · Green outline = our forecast

Conclusion

Built: 6 sources into Bronze, Silver and Gold, 3 use cases served, checks that block bad data, 2 incidents tested. Total cost: €0.

Limits: 4 of 6 sources are simulated.
Next: bring the first SQL objects under Terraform, and reject impossible dates before the live table.

